Units 12, 13
Programming Languages

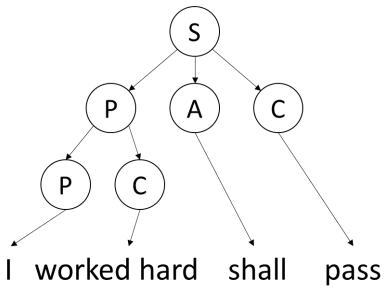
1. Consider the following grammar, where capital symbols are non-terminals, bolded words are terminals, '|' means or, and S is the starting symbol.

S is the starting symbol and bold symbols are terminals:

S → P C | P A C | P C C
A → **shall** | **might** | **have** | **shall have**
P → **i** | **you** | P C
C → **worked hard** | **pass** | **fail**

(a) No. "you might have pass" needs be derived from PAAC, which is not possible to be derived from S. S can go to PC, PAC, or PCC, but then the only way to generate more non-terminals is by P->PC, which does not generate more A that we need.

(b)



(c) Yes. It is ambiguous. For example, PCC can be generated by S->PCC, or by S->PC->PCC.

2. Ans:
Encapsulation:
(a) A way of restricting access to the internal components of an object
(b) Bundling of data with the methods operating on that data.

Polymorphism:
(a) Allows methods calls to be interpreted by the object that receives the call.
(b) Allows different data types to be handled using a uniform interface.

3. Ans:

```
f(1,1).
f(N,SSUM):-
        N>1,
        N1 is N-1,
        f(N1,SSUM1),
        SSUM is SSUM1+N*N.
```